

VISHWAS H K

ANDROID AUTOMOTIVE HMI/ FRAMEWORK/ INFOTAINMENT DEVELOPER/APPLICATION DEVELOPER

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Propelling innovation in next-generation Android-based automotive infotainment by contributing in HMI, middleware, or framework development, with a strong focus on system stability, seamless user experience, and OEM-grade quality.

PROFILE SUMMARY

- Focused **Automotive Android Developer** with **over 4 years of experience** in designing and delivering **infotainment features** on **Android Automotive OS** for EVs and **connected vehicles**, with strong capabilities in **HMI applications, media frameworks, and middleware enhancements**.
- Last worked at **KPIT** for **Jaguar Land Rover (JLR)**; delivered **system-level improvements** for JLR's **next-gen EV media framework** and **IVI user experience**, resolving key issues in **media session management, source switching, and vehicle-state-driven HMI logic**.
- Experienced in building responsive **IVI apps (Media Player, Sentry Mode, Park Assist, MyVehicle, Ambient Lighting)** using **Kotlin, Java, Jetpack Compose, MVVM, and Navigation Component**, adhering to **OEM branding and UX standards**.
- Proficient in integrating core Android components like **CarPropertyManager, AudioManager, and CarAudioService** to enable **real-time media controls, audio focus switching (BT A2DP, USB, FM/AM), and power state transitions**; skilled in **projection technologies** including **Android Auto** and **Apple CarPlay (iAP1/iAP2, Bonjour service)**, ensuring **seamless integration, MFi certification, and performance tuning**.
- Rich experience in **reverse engineering, SAST/DAST testing, malware auditing, secure coding practices, UI instrumentation, and unit testing** to ensure **robust and secure infotainment software**.
- **Previously worked at Mahindra & Mahindra Ltd., contributed extensively to B6 and XEV 9e vehicles** in media applications.
- Proven skills in **IVI app development, framework customization, AOSP feature integration, CAN signal mapping, debugging using Android log tools**, ensuring compliance with **automotive safety and software standards (ASIL, AUTOSAR)**.

CORE COMPETENCIES

Android Automotive OS (AAOS)	Android Auto & Apple CarPlay Integration	Instrumentation & UI Testing
HMI Application Development	AOSP & HAL Customization	Linux System & Log Debugging
Android Framework & Middleware Customization	Multimedia Framework	Agile Development & CI/CD Tools
Media & Audio Stack Integration	Binder IPC, AIDL, HIDL	OEM & Tier-1 Collaboration
CAN Protocol & Signal Analysis	Secure Android Development	Android Debugging & Crash Analysis

AWARDS & ACHIEVEMENTS

- **Best Incoming Fresher Award:** Received at Cognizant Technology Solutions for outstanding performance as a new joiner.
- **Performance Excellence Service Delivery Award:** Awarded at Mahindra & Mahindra for delivering high-quality IVI application features.
- **Spot Award:** Recognised at KPIT Technologies for impactful work on Android Automotive HMI and framework components.
- **Manager's Discretionary Award:** Acknowledged at KPIT Technologies in recognition of consistent technical contributions and delivery excellence.

TECHNICAL SKILLS

- **Programming Languages & Concurrency:** Core Java, Java 8, Kotlin, Kotlin Coroutines, C++, Socket Programming, Multithreading, JNI
- **Android Application & HMI Development:** Android Application Development, Jetpack Compose, MVVM Architecture, Room Database, Navigation Component, Camera App, Instrumentation Testing (UI Testing), Unit Testing, Figma (UI/UX), Flutter, Third-Party Libraries & APIs
- **Android Framework & Middleware:** Android SDK, Android Framework, Android Automotive OS, AOSP Architecture, Binder IPC, AIDL / HIDL, SELinux Policies, Application Security (SAST/DAST), Android Telephony, Linux
- **Infotainment & Automotive Systems:** Automotive Infotainment, Multimedia Infotainment, Android Auto, Apple CarPlay, Projection Technology, ADAS, Telematics, Multimedia Framework
- **Embedded & Connectivity Tools:** AUTOSAR, CANoe, Vehicle Spy, RAD-Gigastar, Connectivity, Hardware Testing, GIT
- **DevOps, Testing & Project Tools:** JIRA, JIRA Xray, GitHub, Penetration Testing

WORK EXPERIENCE

KPIT Technologies Limited, Bangalore, India
Oct'24 – Jun'25 | Technical Leader (Grade D)

Highlights:

- Led development efforts integrating Android's CarPropertyManager and AudioManager services to dynamically manage real-time display and volume settings based on vehicle state transitions.
- Enhanced MediaSession and PlaybackService components to support speed-sensitive media controls and seamless audio source switching logic.
- Engineered audio routing and focus management workflow between Bluetooth A2DP, USB, and Radio sources by leveraging AudioManager and CarAudioService interfaces.
- Diagnosed and resolved complex issues related to metadata and album art propagation between MediaBrowserService and custom HMI components.

- Conducted rigorous testing and validation of media playback behaviors during power state events (e.g., ACC OFF, ignition ON, mute/unmute) to ensure robust framework stability and user experience consistency.
- Mentored junior engineers in debugging techniques, Android HAL integration, and system-level testing to elevate team competency and reduce issue resolution time.
- Collaborated cross-functionally with HMI, Hardware, and QA teams to define feature requirements, improve development timelines, and ensure end-to-end system alignment.

Mahindra & Mahindra Ltd. – Software Defined Vehicle Center (SDVC)

Oct'23 – Oct'24 | Senior Software Engineer (L7 – Operational)

Highlights:

- Contributed to the development of Human-Machine Interface (HMI) systems for Mahindra vehicles, with a focus on digital instrument clusters and In-Vehicle Infotainment (IVI) platforms using **Kotlin** and **Java**.
- Played a key role in creating and refining multiple IVI applications such as **Sentry Mode**, **Park App**, **MyVehicle**, and **Ambient Lighting**, delivering enhanced user interaction and vehicle security features.
- Advanced the IVI media experience by implementing **multi-source audio/video input management**, **ExoPlayer integration**, **video playback with screenshot support**, and intuitive **seekbar navigation**.
- Conducted comprehensive **UI and instrumentation testing** to verify app performance, detect UI inconsistencies, and improve responsiveness across different infotainment modules.
- Managed testing workflow in **JIRA**, designing structured test cases that ensured robust validation of app features and interfaces.
- Provided ongoing support in debugging and issue resolution for both core and auxiliary IVI apps, contributing to consistent system performance and end-user satisfaction.
- Administered **hardware-level diagnostics** using tools like **CANoe** and **Vehicle Spy**, validating **Controller Area Network (CAN)** messages and **Electronic Control Unit (ECU)** behaviors as part of vehicle-level system checks.

Capgemini Engineering (R&D Subsidiary of Capgemini Group), Bangalore, India

Apr'23 – Oct'23 | Software Engineering Lead (Professional I / B1)

Highlights:

- Led the design and development of smartphone connectivity and projection technologies (iAP1, iAP2, Apple CarPlay) for automotive infotainment systems, focusing on seamless Apple device integration.
- Managed a global team, implementing Apple connectivity solutions and leading the end-to-end **MFi certification** process, including performance tuning and compliance validation.
- Administered infotainment system development within AOSP-based Android platforms, including wireless protocol integration and multi-application management.
- Supervised the Connectivity domain for smartphone–vehicle integration, with hands-on work in Apple CarPlay Control, Bonjour service, and cross-platform interoperability in connected car environments.

Cognizant Technology Solutions (CTS), Bangalore, India

Dec'20 – Apr'23 | Technical Lead – Application Development & Android Security Lead

Highlights:

- Led Android security operations involving reverse engineering, static/dynamic code analysis, and de-obfuscation to uncover vulnerabilities and reinforce app security.
- Executed comprehensive SAST and DAST testing to mitigate risks from spyware, phishing, ransomware, trojans, rooting exploits, backdoors, and other mobile malware.
- Conducted internal and external vulnerability assessments, engineered detection routines through internal pipelines, and performed data-driven threat analysis.
- Delivered advanced malware analysis and threat classification for Android; audited 10,000+ applications across MuWS and PSL tracks, acting as a Subject Matter Expert (SME).

PUBLICATION

- **Trust Management for Service-Oriented IoT Systems**

Publisher: ACM Digital Library

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PROJECTS UNDERTAKEN

- *KPIT Technologies: Jaguar Land Rover (JLR) (Oct'24 – Jun'25)*
- *Mahindra & Mahindra: Mahindra Electric SUVs HMI IVI Applications (BE.05, BE.07, BE. RALL-E, XUV.e8, XUV.e9) (Oct'23 – Oct'24)*
- *CapGemini: HARMAN (Toyota TAS700) (Apr'23 – Oct'23)*
- *Cognizant: Google Play Protect (G.P.P) (Dec'20 – Apr'23)*

EDUCATION

- **M.Tech. in Computer Science & Engineering**
University Visvesvaraya College of Engineering, Bangalore | 2020 | CGPA: 8.93
- **B.E. in Information Science and Engineering**
Bangalore Institute of Technology (BIT), Bangalore | 2018 | CGPA: 7.23